

TETOUAN, Morocco

WMO: 603180

Lat: 35.594N

Lon: 5.320W

Elev: 3

StdP: 101.29

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.0	7.2	0.9	4.0	13.1	2.2	4.4	12.2	13.4	15.1	12.4	15.3	2.6	220	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	33.4	20.1	31.6	20.2	30.0	20.2	24.1	27.5	23.5	26.8	23.0	26.2	5.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	17.9	25.8	22.3	17.0	25.3	21.9	16.6	25.1	72.3	27.2	70.0	26.9	67.9	26.4	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.4	9.4	3.1	37.1	1.9	2.5	1.7	38.9	0.6	40.3	-0.5	41.7	-1.9	43.6		
			2.2	25.0	1.5	1.3	1.1	25.9	0.3	26.6	-0.6	27.3	-1.7	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	13.8	14.0	15.4	16.9	19.3	22.9	25.2	25.9	23.5	20.5	16.9	14.6
	DBStd	4.72	2.17	2.12	2.02	1.84	2.12	1.87	1.95	1.79	1.96	2.02	2.45	2.18
	HDD10.0	2	1	1	0	0	0	0	0	0	0	0	0	1
	HDD18.3	591	139	121	92	49	13	0	0	0	5	56	117	
	CDD10.0	3332	120	114	169	208	290	386	472	493	406	325	206	142
	CDD18.3	880	1	0	2	7	44	137	214	234	157	72	12	1
	CDH23.3	5598	0	1	13	23	158	738	1722	1922	821	186	14	1
	CDH26.7	1603	0	0	1	2	24	172	577	629	173	24	1	0
Wind	WSAvg	4.5	4.6	5.0	5.0	4.9	4.5	4.8	4.5	4.2	4.2	3.7	4.5	4.3
Precipitation	PrecAvg	684	103	96	80	66	36	13	1	1	20	67	93	130
	PrecMax	1297	454	359	320	201	125	105	8	7	65	209	275	455
	PrecMin	272	0	0	1	0	2	0	0	0	0	4	1	0
	PrecStd	252	100	86	63	50	32	21	2	2	21	58	77	100
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.0	25.1	25.9	29.3	33.0	36.2	36.1	33.1	29.8	25.1	21.9
		MCWB	14.4	14.8	15.5	16.4	18.5	20.2	20.6	19.6	19.5	18.6	16.8	15.0
	2%	DB	19.7	20.0	22.0	23.5	26.8	30.2	33.4	33.6	30.1	26.9	23.2	20.1
		MCWB	14.3	14.1	15.1	16.4	18.0	19.4	20.3	20.1	19.9	18.9	17.3	14.9
	5%	DB	18.5	18.8	20.4	21.9	25.0	28.8	31.4	31.5	28.5	25.2	22.0	19.0
		MCWB	13.9	13.7	14.8	15.6	17.7	19.2	20.2	20.4	20.0	19.1	16.9	14.8
	10%	DB	17.4	17.7	19.0	20.6	23.3	27.0	29.8	29.9	27.1	24.1	20.9	18.1
		MCWB	13.5	13.4	14.5	15.5	17.4	19.5	20.4	20.7	20.6	19.1	16.4	14.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	16.1	17.4	18.6	20.8	23.0	24.7	25.3	23.6	21.8	19.7	17.0
		MCDB	19.0	20.1	21.5	23.8	25.4	27.3	28.9	28.2	27.1	25.3	23.1	19.5
	2%	WB	15.4	15.2	16.4	17.3	19.7	22.2	23.9	24.5	22.9	20.9	18.5	16.1
		MCDB	17.9	18.0	20.0	21.3	23.9	26.2	27.8	27.5	25.9	23.8	21.6	18.7
	5%	WB	14.8	14.7	15.6	16.7	18.9	21.5	23.2	23.8	22.4	20.4	17.7	15.4
		MCDB	17.4	17.3	18.9	20.3	22.7	25.4	26.9	26.8	25.5	23.2	20.6	18.1
	10%	WB	14.1	14.2	15.1	16.0	18.3	20.9	22.6	23.4	21.9	19.7	17.1	14.8
		MCDB	16.9	16.8	18.0	19.4	21.8	24.6	26.2	26.3	24.9	22.5	19.9	17.5
Mean Daily Temperature Range	5% DB	MDBR	6.9	6.9	6.9	6.9	7.2	7.8	8.2	7.8	7.2	6.9	6.9	7.1
		MCDBR	7.8	8.7	9.5	9.1	10.0	10.9	11.6	10.9	9.2	8.7	7.9	7.6
		MCWBR	4.7	5.1	5.4	4.9	4.8	5.2	5.1	5.0	4.0	4.0	4.3	4.6
	5% WB	MCDBR	6.4	7.0	7.6	7.6	7.7	7.7	7.5	6.3	6.1	6.5	6.0	6.7
		MCWBR	4.4	4.8	4.9	4.7	4.3	4.3	4.1	3.4	3.2	3.6	3.7	4.6
Clear-Sky Solar Irradiance	taub	0.336	0.364	0.412	0.410	0.430	0.458	0.472	0.477	0.443	0.407	0.367	0.343	
	taud	2.501	2.414	2.258	2.293	2.264	2.192	2.139	2.138	2.264	2.366	2.455	2.499	
	Ebn at Noon	862	873	855	875	858	831	816	804	814	814	816	829	
	Edh at Noon	83	101	129	130	136	146	153	150	126	105	87	79	
All-Sky Solar Radiation	RadAvg	2.65	3.42	4.49	5.56	6.54	7.51	7.57	6.67	5.23	3.82	2.81	2.36	
	RadStd	0.25	0.42	0.49	0.37	0.50	0.28	0.15	0.26	0.26	0.34	0.23	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	+79

Nomenclature: See separate page